NATIONAL MATHEMATICS CONTEST
PRIMARY-LEVEL PAPER 1.

NAMES: _____ DATE OF BIRTH: _____

SCHOOL/INSTITUTION: _____

CLASS: _____ DISTRICT: _____

INSTRUCTIONS TO CANDIDATES AND SUPERVISORS:

Do not turn this page until you have been instructed to do so. This competition is conducted on the assumption that proper security is maintained. Uganda Mathematical Society reserves the right, should there be evidence of any breach of security or any suspicion, to reject scripts from a particular school/institution.

PLEASE, NOTE CAREFULLY:

1. **WRITE YOUR ANSWERS IN THE SPACES PROVIDED. MORE SPACE IS AVAILABLE AT THE BACK OF EACH PAGE.**
2. Date: Saturday, 18th March, 2017.
3. Time: Exactly 2 hours and 15 Minutes (from 9 : 00am to 11 : 15am)
4. Type:
   (a) This paper consists of two sections and each section carries 50 marks.
   (b) Attempt as many questions as you can. The total marks scored from all the questions you attempt will be your final score.
   (c) No marks will be awarded for an answer if a clear and logical layout of the working is not shown.
5. All diagrams/figures in this contest are not to scale.
6. Immediately the paper is done, the answer script(s) must be forwarded to:

   David Ddumba Walakira,
   Contest Coordinator, Uganda Mathematical Society,
   c/o Department of Mathematics, Makerere University,
   P.O. Box 7062, Kampala.
   Telephone contact: $+256-414-540692$.

   Answer scripts should be received **latest, Monday** 20**th July,** 2017.
7. **ALL** *participants who qualify for Paper* 2 *are cordially invited to the certificate and prize giving ceremony on Saturday* 29*th July,* 2017 *at NOON at Makerere University, Kampala.*
8. National Mathematics Contest 2017 Paper 2 will be done on 17th June, 2017 at various centres.
9. **UGANDA MATHEMATICAL SOCIETY** wishes you success in this year's contest.

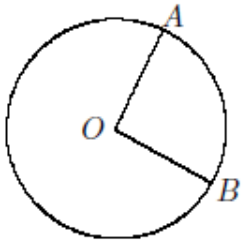
SECTION A : 5 marks each

1. If $T = 8 \times 10000 + 7 \times 1000 + 6 \times 10 + 5 \times 1$, what is T? (*Circle the right answer*).
   A) 8765 *B*) 87065 *C*) 87651 *D*) 87605 *E*) *None of these*

2. The highest temperature on Tuesday was 4 degrees warmer than the highest temperature on Monday. The highest temperature on Wednesday was 6 degrees cooler than the highest temperature on Monday. If the highest temperature on Tuesday was 22 degrees, what was highest temperature on Wednesday?

3. Mary plants ten trees every 3 minutes. If she continues planting at the same rate, how long will it take her to plant 2500 trees?

4. In the circle with centre O, sector AOB represents 20% of the area of the circle, radius $7cm$. Angle AOB = ___ degrees.



5. The pattern of letters A, B, C, D, E is repeated to form the sequence:
   $A, B, C, D, E, A, B, C, D, E, A, B, C, D, E, A, B, \ldots\ldots$ What is the 2204th letter of the sequence?

6. Cathy and John play a two person game in which the winner gains 2 points and the other loses 1 point. If Cathy won exactly 3 games and John had a final score of 5 points, how many games did they play?

7. If point M with coordinates $(3, -2)$ is the midpoint of the line that connects points Y and Z and the coordinates of point Y are $(\frac{1}{2}, -\frac{3}{7})$, then what are the coordinates of Z?

8. Juliet had a $10:00$am appointment 60km from her home. Juliet's average speed was 80km/h for the trip but arrived 20 minutes late for the appointment. At what time did she leave her home?

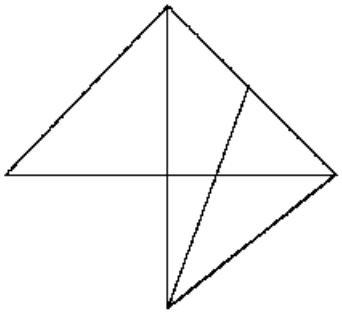
9. Twelve balloons $A, B, C, D, \ldots, L$ are arranged in a circle. Counting clockwise from A, every 3rd balloon is popped. C is the first one popped. This continues until two balloons remain unpopped. Which are the two baloon that remained unpopped?

10. Sarah uses digits $3, 5, 6, 7, 8$ (each once) to form a 3-digit number and a 2-digit number. She then subtracts the two numbers and gets the smallest positive difference. What is this smallest positive difference?

SECTION B : 10 **marks each**

11. Each time a piece of soap is used, its volume decreases by 10%. What is the minimum number of times a new piece of soap would have to be used so that less than one-half of it remains?

12. How many triangles are in the figure?

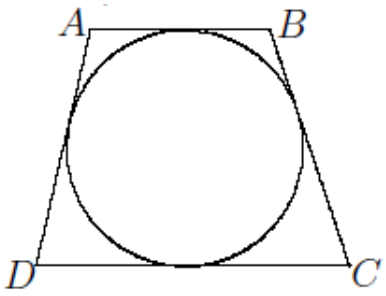


13. Ten points are spaced equally around a circle. How many different chords can be formed by joining any two of these points?

14. In the 4×4 square shown, each row, column and diagonal should contain each of the digits $1, 2, 3, 4$. What is the value of $r + s$?

1			
	2		r
		3	s
		1	

15. A circle is inscribed in trapezoid $ABCD$ with $AD = BC = 25$cm, $AB = 18$cm, $DC = 32$cm.



(a) What is the radius of the circle?

(b) What is the circumference of the circle?

SECTION A : 5 marks each

1.

$$\begin{aligned}(8 \times 10000) + (7 \times 1000) + (6 \times 10) + (5 \times 1) &= 80000 + 7000 + 60 + 5 \\ &= 87065\end{aligned}$$

B

2. The highest temperature on Tuesday is 22 degrees.

   The highest temperature on Monday was $22 - 4 = 18$ degrees.

   But the highest temperature on Wednesday was 6 degrees cooler than was on Wednesday.

   The highest temperature on Wednesday was $18 - 6 = 12$ degrees.

3. Mary plants 1 tree in $\frac{3}{10}$ minutes.

   She plants 2500 trees in $\frac{3}{10} \times 2500 = 12\frac{1}{2}$ hours (or 750 minutes).

4.

$$\angle AOB = \frac{20}{100} \times 360 = 72^\circ$$

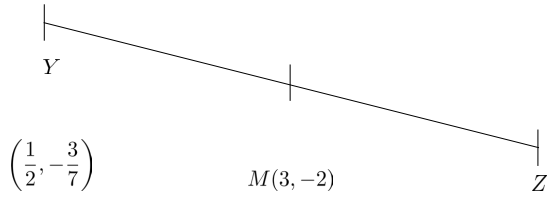
5. The sequence repeats every 5th letter, so will repeat after 2200 (multiple of 5), hence 2204th will be 4th letter D.

6. If Cathy won 3 games, then John lost 3 points, so that he must have had 8 points before loosing in order to have a final score of 5 points.

   If John had 8 points before loosing, then he must have won 4 games.

   If John won 4 games and Cathy 3 games, then there were 7 games played.

7. For the figure



   Let the coordinates of Z be (x_0, y_0), then

$$\begin{aligned}\frac{x_0 + \frac{1}{2}}{2} &= 3 \\ \frac{y_0 + \left(-\frac{3}{7}\right)}{2} &= -2\end{aligned}$$

   This imply $x_0 = 6 - \frac{1}{2} = \frac{11}{2}$ and $y_0 = -4 + \frac{3}{7} = -\frac{25}{7}$. Thus Z has coordinates $\left(\frac{11}{2}, -\frac{25}{7}\right)$

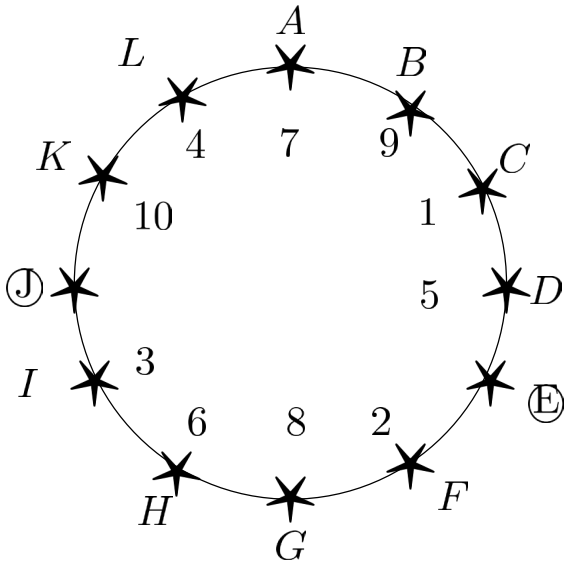
8. At a speed of 80 km/h, Juliet used $\frac{60}{80}$ hr $= \frac{3}{4}$ hr $= 45$ min to cover 60 km.

   But Juliet arrived 20 min late, that is, at 10 : 20 am.

   Thus she set off 45 min before 10 : 20 am, that is at 9 : 35 am.

9. Ten popped baloons in order are

$$C, F, I, L, D, H, A, G, B, K$$



   The unpopped are E and J.

10.

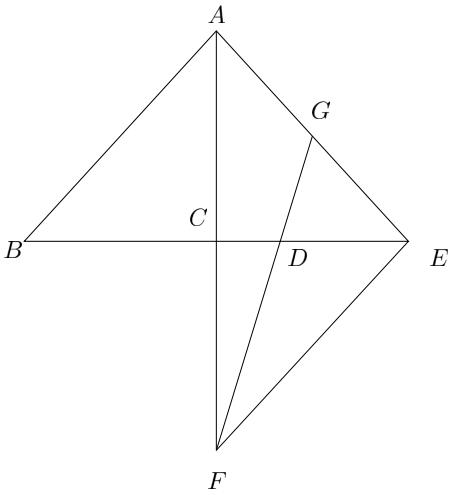
$$\begin{aligned} \text{Smallest postive difference} &= \text{Smallest 3 digit number} - \text{Largest 2 digit number} \\ &= 356 - 87 \\ &= 269 \end{aligned}$$

SECTION B : 10 **marks each**

11. Percentages of volume of soap remaining

Number of times	Amount left
1	90%
2	$90\% \times 90\% = 81\%$
3	$90\% \times 81\% = 72.9\%$
4	$90\% \times 72.9\% = 65.61\%$
5	$90\% \times 65.61\% = 59.05\%$
6	$90\% \times 59.05\% = 53.15\%$
⑦	$90\% \times 53.15\% = 47.84\%$

12. If we name vertices as following



The possible triangles are

A	ABC, ABE, ACE, AFG, AFE	(5)
B	None	(0)
C	CFD, CFE	(2)
G	GDE, GFE	(2)
D	DFE	(1)
E	None	(0)
F	None	(0)

Total is 10.

13. Assume points are numbered 1 to 10.

1 joins with 2 to 10 to make 9 chords.
2 joins with 3 to 10 to make 8 chords.
3 joins with 4 to 10 to make 7 chords.
⋮ ⋮ ⋮ ⋮
9 joins with 10 to make 1 chord.

That is

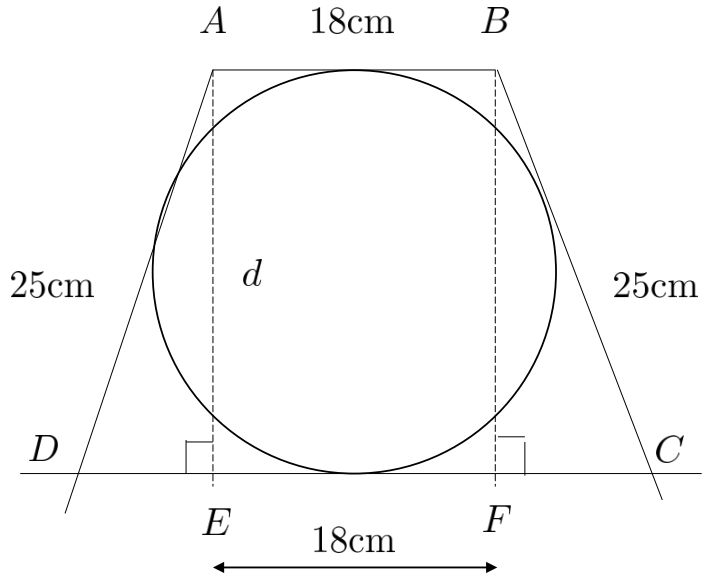
$$9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 45 \text{ chords}$$

14. The square is filled as below

1	4	2	3
3	2	4	1
4	1	3	2
2	3	1	4

$r + s = 1 + 2 = 3$

15. (a) For the figure below,



The sides AB and DC of the trapezoid $ABCD$ are parallel and the angles ADC and angle BCD are equal, hence

$$\overline{DE} = \overline{FC} = \frac{32 - 18}{2} = 7 \text{ cm}$$

The diameter of the circle is $d = \overline{AE} = \overline{BF}$, and

$$d^2 = \overline{AD}^2 - \overline{DE}^2 = 25^2 - 7^2 = 24^2 \quad \Rightarrow \quad d = 24 \text{ cm} \quad \Rightarrow \quad \text{radius } (r) = 12 \text{ cm}$$

(b) The circumference of the circle $= \pi D = 24\pi$ cm